

LEE ELDER

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EDUCATION

BSc (Hons) Computing Science - University of Strathclyde, Glasgow 2023 - 2027
Expected: First Class Honours (current average 70%)

PROJECTS

Grademap - Cross-platform grade tracker and academic planner | grademap.net Nov 2025 - Present

- Designed the calculation engine that works backwards from a target grade to the marks still needed across weighted, partly-marked assessments, turning incomplete data into a prioritised list of what to work on next.
- Guarded schema changes with a migration process that refuses to run unless the target environment is typed out and confirmed, keeping staging and production changes deliberate and separate.
- Built and shipped the full stack alone: a React and TypeScript front end deployed on Vercel, a PostgreSQL database secured with row-level access policies and serverless functions in Deno, and an automated CI/CD pipeline running type checks, linting, tests, and a build on every change.

Momentum - Solo hackathon, GDG on Campus Aberdeen | github.com/energizee/gdgstudyapp Jun 2026

- Shipped a complete working product inside the 6-hour window as a solo entrant against teams of 2-3: a personalised study companion built with Next.js and TypeScript.
- Routed all model calls through one server-side endpoint handling 10 task types, keeping the secret key off the client and centralising caching, timeouts, and fallback behaviour in a single place.
- Treated model output as untrusted input, validating every response against real record identifiers, permitted values, and numeric limits before it reached the application.
- Cut response time roughly 3.5x, from 5.0 to 1.4 seconds, by reducing the model's reasoning budget, and added per-request deadlines with automatic fallback to a faster model after measuring a stalled request run for 352 seconds against a 60-second server limit.

Networked Chat Application - Concurrent client-server system in Java Apr 2026

- Developed a multi-user communication platform over TCP and UDP sockets, supporting real-time messaging, file transfer, and webcam video streaming.
- Implemented concurrency control over shared state using threading and synchronisation primitives, keeping user sessions and persistent chat logs consistent under simultaneous read and write access.
- Selected transport per data type: TCP where ordering and delivery mattered for messages and history, UDP for live video where latency mattered more, handling packet loss and reordering in the video path directly.

Unix Shell - Systems programming in C Apr 2025

- Built a Unix-like shell in C supporting internal and external commands, aliasing and command history; wrote the input parser from scratch and implemented process creation and control via fork/exec alongside direct system calls.

EXPERIENCE

Treasurer - GDG on Campus Strathclyde Sep 2026 - Present

- Own the society's finances: maintain financial records, manage the budget, and track income and expenditure across all events.
- Prepare event budgets and funding applications, handle reimbursements, and coordinate with the Students' Union on financial procedure and compliance.

Shift Leader - Tesco, Glasgow (part-time) Mar 2025 - Present

- Lead shifts on the floor: delegate tasks across the team, train new colleagues, and ensure time-critical routines are completed to schedule.
- Diagnose and resolve faults in in-store systems, handle escalated complaints, and monitor store data to verify routines are completed correctly.

Receptionist - Fraser Suites, Glasgow Jun - Sep 2024

- Managed reservations and administration in the hotel management system, handling check-ins, guest requests, and complaints during peak periods.

TECHNICAL SKILLS

Languages: Java, TypeScript, JavaScript, Python, C, C++, SQL, HTML, CSS

Backend & data: PostgreSQL (schema design, row-level security), Supabase, Deno, REST APIs, TCP/UDP sockets, concurrency

Frameworks & tools: React, Next.js, Node.js, Electron, React Native, Zustand, Git, GitHub Actions (CI/CD), Vercel, Linux, Vitest

ACHIEVEMENTS AND INTERESTS

Young STEM Leader (2023): Designed and delivered science lessons for primary school pupils, including one on programming BBC micro:bits.

Half Marathon: Completed the 2026 Edinburgh Marathon Festival Half Marathon, raising money for Scottish Autism.

Interests: Brazilian Jiu Jitsu, bass guitar, gym, chess